Boundary-Guided Replay: A Mechanism Study of Success–Failure Boundary Learning

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Abstract

Policies trained on nominal starts often fail under small deviations from the states encountered during training. Existing curriculum and level-replay methods prioritize difficult or high-regret levels, but they do not identify where behavior transitions from likely success to likely failure. We introduce Boundary-Guided Replay (BGR), a replay method for resettable or replayable sequential decision settings. For each replayable state, BGR estimates a recovery curve over perturbation scale, derives a critical radius, and samples training perturbations near this state-conditioned boundary. The paper is scoped as a mechanism study, not as a completed robotics or broad-benchmark result. The clearest effect is in a procedural grid-margin benchmark, where BGR improves recovery AUC over uniform replay in both the 30-seed full-baseline run and a held-out 30-seed replication (pooled 0.4342 vs. 0.3965). Synthetic and robot-suffix results are smaller and are reported as scoped support: the coverage-aware suffix result is manipulation-style support rather than broad robustness evidence, improving final object recovery AUC (0.4971 vs. 0.4856), clean success, transfer recovery AUC, and learning-curve area, while still trailing uniform on median critical radius. Standard-environment and OpenVLA/LIBERO audits are negative for promotion as broad robotics evidence. They show that the interface can measure learned-policy brittleness and build matched fine-tuning datasets, but current adaptation does not establish stable gains over the official checkpoint.

1 Introduction

A policy can solve nominal starts and still be brittle: a small change in layout, object pose, or trajectory prefix may move it outside its recovery basin. Standard success metrics hide this geometry. We ask a deliberately narrower question: for each replayable state, how far can the current policy be perturbed before success becomes failure, and can replay near that boundary expand the local margin?

Existing replay and curriculum methods rank examples by difficulty, temporal-difference error, regret, novelty, or failure. These scalar signals are useful, but they do not explicitly model the local response curve around a state or decide which perturbation radius should be trained once a state is selected. BGR instead estimates a state-conditioned recovery curve and trains near its success–failure transition. The empirical question is therefore not whether another priority score can win small paired comparisons; it is whether boundary-local radius selection adds anything beyond uniform replay, fixed radii, failure mining, and loss-priority proxies.

Our contributions are:

- a domain-agnostic replay interface based on replayable decision states, perturbation families, recovery curves, critical radii, and recovery AUC;
- an efficient BGR algorithm that actively estimates success–failure boundaries and samples both states and perturbation radii near those boundaries;
- an active-estimation validation showing that boundary-focused probes recover critical radii with small budgets;
- synthetic, procedural grid-margin, and robot-suffix simulator benchmarks showing where boundary replay and coverage-aware boundary replay expand recovery margins, including a 30-seed grid ablation where state-priority replay with uniform radii loses the effect;
- negative standard-environment and OpenVLA/LIBERO audits that define the practical scope of the evidence and separate recovery-margin measurement from robotics fine-tuning claims.

The claim is intentionally scoped. BGR is not a universal robustness guarantee, broad independent-benchmark result, or completed robotics fine-tuning method; standard-environment probes are negative or exploratory, and OpenVLA/LIBERO is audit evidence. BGR is a replay rule for settings with revisitable states, feasible perturbations, and trainable recovery curves. Controlled experiments test the mechanism under paired seeds and ablations; suffix and OpenVLA/LIBERO studies test interface transfer and expose where learned-policy claims still need stronger evidence.

2 Related Work

BGR is closest to curriculum learning and level replay. Prioritized Level Replay samples levels by value-loss learning-potential proxies (Jiang, Grefenstette, and Rocktäschel 2021); PAIRED and ACCEL use regret and curriculum-frontier objectives (Dennis et al. 2020; Parker-Holder et al.

2022). The novelty claim is narrow and falsifiable: BGR changes the target from a level or transition score to an estimated curve crossing and the replay action from revisiting a hard state to choosing a radius near that crossing. If state-priority replay with uniform radii or loss-priority replay matches BGR, the boundary claim collapses to ordinary hard-example replay. The radius-level ablation is therefore a negative control: state priority is held fixed and only the radius rule changes. Drawing radii uniformly removes the grid-margin gain. BGR also differs from broad curriculum schedules (Narvekar et al. 2020): progress means recovery-margin expansion around each state, not global task difficulty.

Prioritized experience replay ranks transitions by temporal-difference error (Schaul et al. 2016), and active learning selects informative labels under a query budget (Settles 2009). BGR borrows the budgeted-query spirit but estimates a success-probability curve around a replay state, then trains at a state-conditioned radius. It is also distinct from adversarial-example training (Goodfellow, Shlens, and Szegedy 2015; Madry et al. 2018): BGR is not an adversarial-training objective and does not solve a worst-case inner maximization; it samples feasible scales near the observed boundary to expand local margin.

BGR also connects to robustness training and robotics generalization. Domain randomization broadens training distributions (Tobin et al. 2017), but fixed perturbation distributions can oversample states that are already easy or infeasible. BGR estimates each state's success–failure crossing and trains near that radius. Our procedural experiments use resettable grid states in the style of lightweight grid benchmarks (Chevalier-Boisvert et al. 2023); the suffix study instantiates the interface for manipulation-style states related to LIBERO (Liu et al. 2023) and OpenVLA (Kim et al. 2024).

3 Problem Setting

Let π_θ be a sequential decision policy and let $Y(\tau) \in \{0,1\}$ denote terminal success for rollout τ. A replayable decision state $x \in \mathcal{X}$ is any object from which evaluation or training can restart, such as an environment seed, simulator checkpoint, task prefix, or robot suffix state.

For each x, a domain-valid perturbation family applies $\tilde{x} = T(x,\delta)$ with $\delta \sim \mathcal{P}_x(\sigma)$, where $\sigma \geq 0$ is a perturbation magnitude. The perturbation family is part of the benchmark definition: grid perturbations preserve free cells and shortest-path reachability, suffix perturbations preserve teacher feasibility, and OpenVLA observation perturbations preserve the task instruction and simulator state while modifying the visual input stream. The recovery curve is

$$R_\pi(x,\sigma) = \Pr_{\delta,\tau}[Y(\tau) = 1 \mid \tau \sim \pi(\cdot \mid T(x,\delta))]. \quad (1)$$

For threshold α, the critical radius is

$$r_{\pi,\alpha}(x) = \sup\{\sigma : R_\pi(x,\sigma) \geq \alpha R_\pi(x,0)\}. \quad (2)$$

We evaluate robustness with recovery AUC,

$$\mathrm{RAUC}_\pi(x) = \frac{1}{\sigma_{\max}} \int_0^{\sigma_{\max}} R_\pi(x,\sigma) d\sigma. \quad (3)$$

Study	Replay state	Perturbation	Witness/eval
Synthetic	latent recovery state	scalar radius	dense curve
Grid-margin	mid-path grid state	valid radius shifts	reachability
Robot suffix	demo suffix phase	object/EEF offsets	teacher feasibility
OpenVLA audit	LIBERO state	object/visual changes	simulator success

Table 1: Instantiations of the BGR interface. The first three rows are training experiments; the OpenVLA row is an audit of recovery curves and data plumbing, not a fine-tuning claim.

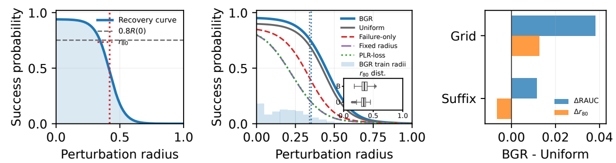

Figure 1: Geometric intuition for BGR, empirical trace, and metric cross-checks. Left: a recovery curve maps perturbation radius to success probability, and the critical radius is the success–failure crossing relative to clean success. Middle: a deterministic seed-0 grid-margin replay trace overlays final recovery curves for BGR, uniform, failure-only, fixed-radius, and PLR-loss replay with the BGR training-radius histogram and an inset showing BGR/uniform critical-radius distributions. Right: paired mean differences against uniform expose the metric tension: grid improves both RAUC and median r_{80}, while the suffix variant improves RAUC but reduces median r_{80}.

BGR depends on a feasibility witness $F(x,\sigma)$ estimating whether perturbed states remain meaningful and solvable by a reference process. This is a real interface assumption, not free supervision or a learned success oracle: without it, boundary replay can waste updates on invalid perturbations or inflate margins with impossible states. Our witnesses are exact reachability checks, teacher/demo suffix models, simulator validity checks, or observed clean OpenVLA success. When no reliable witness is available, BGR should be treated as an audit tool or not applied. We only promote settings where the witness is exact or stress-tested; the suffix stress sweep varies teacher quality to test one witness-fidelity failure mode. A 30-seed grid-margin witness diagnostic keeps exact-witness valid accepted samples at 1.0000; symmetric 10%/20% witness noise preserves valid-accept rates at 1.0000/0.9999 but lowers true-boundary recall to 0.9001/0.7980.

4 Boundary-Guided Replay

BGR maintains a buffer of replayable states. For each state, it probes perturbation radii, fits a monotone recovery curve, estimates a critical radius, and assigns replay priority to states whose boundaries are feasible, sharp, uncertain, and near a trainable target band. The method therefore separates three decisions: whether a state is valid enough to replay, which replay state is currently informative, and which perturbation radius lies near that state's success–failure transition.

Boundary-Guided Replay
Initialize replay buffer $\mathcal{B}$ with replayable states.
For each $x \in \mathcal{B}$, probe $\sigma \in \{0, \sigma_{\min}, \sigma_{\text{mid}}, \sigma_{\max}\}$, fit monotone $\hat{R}(x, \sigma)$, and estimate $\hat{r}_\alpha(x)$.
For each training iteration, sample $x \sim q_{\mathcal{B}}(x)$ proportional to BGR priority, sample σ from a mixture centered at $\hat{r}_\alpha(x)$, apply $\delta \sim \mathcal{P}_x(\sigma)$, collect rollout or teacher recovery data, update π_θ, and periodically refresh stale curve estimates.

Method box: BGR training loop without domain-specific learner details.

The priority for a record is a product of gates and soft utilities:

$$P(x) = G(x) U_{\text{boundary}}(x) U_{\text{sharp}}(x)^\lambda U_{\text{unc}}(x)^\beta U_{\text{stale}}(x)^\eta. \tag{4}$$

Here $G(x)$ gates out states with low clean success or low witness feasibility, U_{boundary} prefers critical radii in a trainable band, U_{sharp} prefers informative transitions, U_{unc} refreshes uncertain curve estimates, and U_{stale} revisits old records as the policy changes. We mix this distribution with uniform replay to prevent starvation. For the radius-level decision, BGR samples from a mixture centered at $\hat{r}_\alpha(x)$ with clean, easier, harder, and broad-radius coverage components; ablations test which part is responsible for the gains.

Two separations are important. First, BGR does not equate failure with usefulness: perturbations far beyond the current recovery radius can be valid but uninformative for margin expansion, while clean or near-clean states may already be saturated. Second, BGR separates replay-state priority from perturbation-radius choice. A state can be worth revisiting because its boundary is uncertain or sharp, but the training sample still needs to land near the recoverable transition rather than at an arbitrary radius. This distinction is what the radius-level ablation tests.

The following local calculation is a design rationale for the radius sampler and for the radius-level ablation, not a convergence result, global robustness theorem, or global margin-expansion guarantee.

Local boundary intuition. Fix a replay state x, current policy π, and critical radius $r = r_{\pi,\alpha}(x)$. Let $\Delta_x(\sigma)$ denote the expected first-order change in held-out RAUC after one small update from a perturbed rollout at radius σ, holding the state distribution, learner linearization, and feasibility witness fixed. Assume this local gain depends only on distance to the current boundary, $\Delta_x(\sigma) = k_x(|\sigma - r|)$, for a nonincreasing kernel k_x. If a boundary sampler $p_b(\sigma \mid x)$ induces a distance-to-boundary random variable $D_b = |\sigma - r|$ satisfying $\Pr[D_b > t] \le \Pr[D_0 > t]$ for every t relative to a baseline sampler's distance D_0, then

$$\mathbb{E}_{p_b}[k_x(|\sigma - r|)] \ge \mathbb{E}_{p_0}[k_x(|\sigma - r|)]. \tag{5}$$

The only mathematical assumption is first-order stochastic dominance over distance to the boundary plus monotonicity of the local gain kernel. By the quantile coupling for stochastic orders (Shaked and Shanthikumar 2007), there exists a coupling with $D'_b \le D'_0$ almost surely and with the same marginals as D_b, D_0; monotonicity of k_x then gives $\mathbb{E}[k_x(D'_b)] \ge \mathbb{E}[k_x(D'_0)]$, which is the stated inequality. A local margin-shift corollary makes the implication concrete: if one small update at radius σ shifts the current critical radius right by at least $c\, k_x(|\sigma - r|)$ and never shifts it left, then the expected one-step critical-radius gain under p_b is at least that under p_0, and any local RAUC lower bound proportional to this shift inherits the same ordering. This is still only a one-step, fixed-state statement. BGR uses boundary-centered sampling as a default but retains clean, easier, harder, broad-radius, and uniform components because real learners can violate the local model through saturation, noisy labels, witness errors, or unstable optimization.

Evidence tier	Supports	Does not support
Synthetic	sampler/estimator sanity	independent benchmark win
Grid-margin	boundary-radius mechanism	broad environment dominance
Robot suffix	manipulation-style interface	final robotics fine-tuning claim
Gym/OpenVLA	scope and audit checks	promoted positive result

Table 2: Evidence contract used throughout the experiments. The central claim is mechanism-level margin expansion when recovery curves are directly trainable; independent-benchmark and learned-policy wins remain open requirements rather than hidden claims.

Finite-grid estimator guarantee. For N states, K radii, and m Bernoulli rollouts per state-radius pair, Hoeffding plus a union bound gives $\max_{x,j} |\bar{R}(x, \sigma_j) - R(x, \sigma_j)| \le \epsilon = \sqrt{\log(2NK/\delta)/(2m)}$ with probability $1 - \delta$. If the true crossings have local slope at least γ and grid spacing is at most Δ, then every estimated critical radius is within $\Delta + (1 + \alpha)\epsilon/\gamma$; achieving error $\eta > \Delta$ scales as $m = O(\log(NK/\delta)/(\gamma^2(\eta - \Delta)^2))$. This certifies boundary localization, not learner improvement.

5 Experiments

All experiments are driven by artifact YAML configs and scripts. The anonymous artifact records per-seed CSVs, aggregation code, paired exact sign tests, Slurm launch commands, run ledger files, compute/GPU environment snapshots, and an MIT license; the paper tables are generated from these artifacts rather than hand-edited results.

Table 2 summarizes the claim boundary before the results. The main claim is supported by controlled synthetic and procedural grid-margin experiments where the recovery-margin object is directly trainable and baselines share paired seeds; the grid result is replicated on held-out seeds and pooled over 60 paired comparisons. The robot-suffix simulator is a manipulation-style extension: a coverage-aware BGR variant improves final object RAUC, clean success, transfer RAUC, and AULC over uniform suffix replay across original and held-out sweeps and in four stress regimes, while still trailing uniform on median critical radius. Gym-style standard-environment probes and OpenVLA/LIBERO results are scope checks and learned-policy audits, not promoted positive claims.

5.1 Protocol and Metrics

We report clean success, recovery AUC (RAUC), median r_{80}, and area under the RAUC learning curve (AULC). RAUC and r_{80} are evaluated on dense held-out radius grids rather than on the probes used by BGR during training. RAUC and AULC are author-defined summaries over selected perturbation families and radius ranges, so we do not treat them as stand-alone proof of robustness; clean success and median r_{80} are reported as complementary checks, and disagreements are discussed directly. Synthetic training, estimator validation, and packaged grid full-baseline, ablation, sensitivity, and suffix coverage confirmations use 30 paired seeds; grid learning-curve diagnostics use 15. We treat exact paired sign tests as consistency checks over shared seeds, not as substitutes for effect size: a 30/0 sign pattern can certify direction while the absolute gain remains small. The synthetic and suffix gains are therefore modest positive effects, while the grid-margin mechanism benchmark shows the largest practically visible gap. Five-seed exploratory variants and small pre-promotion standard-environment probes are reported only as scope diagnostics.

Before promoting a new result as positive evidence, we require it to use a fixed protocol, beat uniform replay on final RAUC with a visible effect size, beat fixed-radius, failure-only, and loss-priority baselines, avoid contradictory median-r_{80} behavior or report it as a limitation, and either replicate on held-out seeds or pass a preregistered stress sweep. This rule is why the Gym-style probes below and the OpenVLA adaptation audits are not used to expand the claim.

Baselines are matched to the decisions BGR changes. Uniform replay samples replay states and perturbation radii without recovery-boundary information. Fixed-radius replay uses the same learner and feasibility filters at one configured perturbation scale. Failure-only replay prioritizes states with observed failures without estimating a crossing radius. PLR-loss and loss-priority baselines replace BGR's recovery-curve priority with learner-loss proxies. Clean-only suffix training regularizes on unperturbed suffixes. Paired comparisons share seeds, rollout budgets, evaluation grids, and validity filters.

5.2 Active Boundary Estimation

BGR is useful only if recovery boundaries can be estimated without evaluating dense curves for every replay state. We therefore validate the training-time estimator separately from the replay experiments. For each synthetic state, we compute a dense 201-point reference curve for evaluation only, then compare 17-probe estimators: uniform random radii and active BGR probing initialized with a coarse sweep and then focused near the current critical-radius estimate.

Table 5 shows over 30 paired seeds that active BGR finds the boundary within ± 0.08 radius units more often than uniform probing (0.670 vs. 0.595) and lowers mean r_{80} error (0.081 vs. 0.105). The paired consistency is reported in Table 4; the practical point is the fixed-budget effect size, not the p-value. This result does not claim that active probing dominates every fixed grid in this simple one-dimensional setting; rather, it verifies that BGR's boundary estimates are usable at a small fixed rollout budget before final dense-curve evaluation.

5.3 Synthetic Recovery-Margin Benchmark

The synthetic benchmark creates replayable states with latent feasible radii and policy recovery margins. Success probability follows a sigmoid recovery curve, and training updates near the current margin expand that margin most efficiently. This benchmark is deliberately constructed to expose the mechanism BGR targets; it is a sanity check for the estimator and sampler before higher-cost sequential or robotics runs, not independent benchmark evidence.

In the 30-seed run, BGR gives a modest final recovery-AUC gain over uniform replay (0.3716 vs. 0.3633; mean difference 0.0084), a larger learning-curve gain (AULC 0.3149 vs. 0.2984), and similar clean success (0.8899 vs. 0.8838). The paired signs are consistent (Table 4), but the practical claim is limited: explicit boundary-radius sampling helps when the benchmark contains learnable success–failure transitions. Fixed-radius, failure-only, and PLR-loss proxy baselines underperform substantially in this controlled setting.

5.4 Procedural Grid-Margin Study

We instantiate replayable states in procedural grid environments with resettable seeds and mid-path states. Generated obstacle maps define feasible perturbation neighborhoods through shortest-path reachability, while each replay state has a policy-specific recovery margin. This benchmark isolates whether BGR can expand state-conditioned recovery curves in a non-robotic sequential decision setting.

In a completed 30-seed paired grid-margin full-baseline run, BGR gives the largest absolute gap in the paper: final RAUC 0.4344 vs. 0.3963, RAUC learning-curve area 0.3526 vs. 0.3132, median r_{80} 0.3447 vs. 0.3321, and clean success 0.9455 vs. 0.8939. A held-out seeds 30–59 BGR-vs-uniform replication gives 0.4340 vs. 0.3967 RAUC, also with 30/0 wins. Pooling original and held-out grid sweeps gives 0.4342 vs. 0.3965 RAUC with 60/0 paired wins. The same sweep shows that failure-only replay (RAUC 0.2909), PLR-loss proxy replay (0.2109), and fixed-radius perturbation training (0.2097) substantially underperform BGR (0.4344).

Figure 2 shows the stored 15-seed learning curves rather than only the endpoint table: BGR separates from uniform early and remains ahead through the final checkpoint over steps 30–300. The endpoint gap is about 0.038 RAUC, which is the paper's clearest practical effect but still a scoped mechanism result rather than broad benchmark dominance. The 30-seed sensitivity tables give the same pattern: target margins 0.26–0.54 yield RAUC 0.416–0.443 vs. uniform 0.396; obstacle regimes yield BGR RAUC 0.434–0.435 vs. uniform 0.396; and geometry stresses give stress RAUC 0.362–0.457 vs. uniform 0.323–0.430. The learning-rate table preserves the high-rate final-RAUC caveat. The reported 0.38 margin is not the best tuned value.

A learning-rate sweep is a scope diagnostic: BGR wins final RAUC at 0.015 and 0.030 (0.382 vs. 0.320; 0.434 vs. 0.396) and AULC at all three rates, but high learning rate flips final RAUC to uniform (0.485 vs. 0.491). Thus neither failure mining nor static perturbation scales recover

Benchmark	Method	Clean	RAUC	Median r_{80}	AULC
Synthetic	BGR	0.890±0.001	0.372±0.001	0.286±0.002	0.315±0.001
Synthetic	Uniform	0.884±0.001	0.363±0.001	0.282±0.002	0.298±0.001
Synthetic	Fixed	0.821±0.001	0.230±0.001	0.131±0.001	0.229±0.001
Synthetic	Failure	0.817±0.001	0.233±0.001	0.135±0.001	0.231±0.001
Synthetic	Loss-priority	0.819±0.001	0.229±0.001	0.134±0.001	0.229±0.001
GridMargin	BGR	0.945±0.000	0.434±0.000	0.345±0.001	0.353±0.000
GridMargin	Uniform	0.894±0.000	0.396±0.000	0.332±0.001	0.313±0.000
RobotSuffix	BGR-Coverage	0.864±0.001	0.497±0.001	0.498±0.001	0.382±0.001
RobotSuffix	Uniform	0.836±0.001	0.485±0.001	0.505±0.001	0.371±0.001

Table 3: Results generated from anonymous artifact CSVs. Synthetic, grid-margin full-baseline, and robot-suffix coverage comparisons use 30 paired seeds, with held-out 30-seed BGR-vs-uniform replications reported in text for grid and suffix. BGR is strongest in the synthetic and procedural grid-margin settings. In the lightweight robot-suffix simulator, the coverage-aware BGR-Suffix variant improves clean success, final object RAUC, transfer RAUC, and AULC over uniform suffix replay, but uniform retains a higher median critical radius.

Comparison	Metric	n	Mean diff.	95% CI	W/L/T	p
Synthetic vs uniform	final RAUC	30	0.0083	[0.0059, 0.0107]	29/1/0	< 0.0001
Grid vs uniform	final RAUC	30	0.0381	[0.0372, 0.0390]	30/0/0	< 0.0001
Suffix RAUC vs clean-only	final RAUC	30	0.2338	[0.2327, 0.2350]	30/0/0	< 0.0001
Suffix RAUC vs fixed	final RAUC	30	0.3386	[0.3373, 0.3399]	30/0/0	< 0.0001
Suffix RAUC vs failure-only	final RAUC	30	0.0741	[0.0731, 0.0750]	30/0/0	< 0.0001
Suffix RAUC vs loss-priority	final RAUC	30	0.3275	[0.3262, 0.3288]	30/0/0	< 0.0001
Suffix RAUC vs uniform	final RAUC	30	0.0116	[0.0109, 0.0122]	30/0/0	< 0.0001
Suffix transfer vs uniform	transfer RAUC	30	0.0061	[0.0054, 0.0068]	30/0/0	< 0.0001
Suffix AULC vs uniform	RAUC AULC	30	0.0115	[0.0108, 0.0122]	30/0/0	< 0.0001
Estimator 30-seed	boundary hit	30	0.0751	[0.0646, 0.0857]	30/0/0	< 0.0001

Table 4: Selected paired effect sizes for the central claims, including the completed 30-seed synthetic, estimator, grid, and suffix full-baseline confirmations. Mean differences and 95% confidence intervals are the primary quantities; W/L/T and exact sign-test p values report consistency of direction over paired seeds. Synthetic, estimator, grid full-baseline, and suffix coverage-full comparisons use 30 paired seeds.

Estimator	r_{80} MAE	RAUC MAE	Hit rate
Active BGR	0.081±0.001	0.065±0.000	0.670±0.004
Uniform	0.105±0.001	0.066±0.000	0.595±0.004

Table 5: Probe-efficiency validation on synthetic recovery curves. Both methods use 17 Bernoulli probes per state and are evaluated against dense 201-point curves. Active BGR improves boundary-hit rate over uniform probing at the same rollout budget, while keeping critical-radius and RAUC error comparable.

the same margin-expansion behavior, while aggressive optimization can saturate uniform final RAUC. A separate tabular grid-policy setup either saturates under broad replay or undertrains BGR relative to fixed-radius oracle imitation; we therefore keep the main procedural claim on the margin-expansion benchmark.

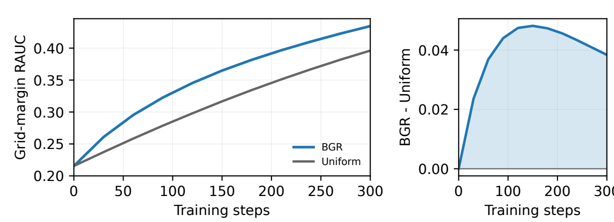

Figure 2: Grid-margin RAUC learning curve from the stored 15-seed run. Shaded bands show standard error. The right panel plots the paired mean gap, making the effect size visible without relying on significance notation.

Method	Clean	RAUC	Median r_{80}	AULC
BGR	0.945±0.000	0.434±0.000	0.345±0.001	0.353±0.000
Uniform	0.894±0.000	0.396±0.000	0.332±0.001	0.313±0.000
Failure-only	0.845±0.000	0.291±0.000	0.231±0.001	0.254±0.000
Loss-priority	0.797±0.001	0.211±0.000	0.137±0.001	0.213±0.000
Fixed radius	0.794±0.001	0.210±0.000	0.136±0.001	0.213±0.000

Table 6: Thirty-seed procedural grid-margin full-baseline

comparison. Boundary-centered replay is the only method that both preserves clean success and expands recovery margins.

Method	Clean	RAUC	Median r_{80}	AULC
BGR	0.945±0.000	0.434±0.000	0.345±0.001	0.353±0.000
No uncertainty	0.946±0.000	0.435±0.000	0.344±0.001	0.353±0.000
No sharpness	0.946±0.000	0.435±0.000	0.344±0.001	0.353±0.000
BGR state + uniform radius	0.891±0.000	0.382±0.000	0.322±0.001	0.305±0.000
Uniform state + uniform radius	0.894±0.000	0.396±0.000	0.332±0.001	0.313±0.000

Table 7: Grid-margin ablations over 30 seeds. Boundary-centered radius sampling is the important ingredient in this benchmark: replacing it with uniform radii drops RAUC below uniform replay, while removing uncertainty or sharpness weights has little effect.

Table 7 isolates the replay decisions inside BGR. The uncertainty and sharpness factors are not decisive in this benchmark, but the radius-level boundary rule is: keeping BGR state priority while sampling perturbation radii uniformly reduces RAUC from 0.434 to 0.382 and AULC from 0.353 to 0.305, even below uniform replay's 0.396 RAUC. A held-out seeds 30–59 replication gives 0.434 vs. 0.382 RAUC and 0.352 vs. 0.305 AULC for BGR over the uniform-radius ablation, again with 30/0 paired wins; uniform-radius remains below uniform replay (0.382 vs. 0.397 RAUC). This is the strongest novelty evidence in the paper: the gain is not explained by hard-state prioritization alone, because state priority is held fixed and only the radius rule changes.

5.5 Robot Suffix Study

BGR-Suffix treats a robot demonstration suffix as a replayable state. The lightweight simulator models language-conditioned manipulation families, suffix phase, object-pose and end-effector perturbations, teacher feasibility, and clean-demo regularization. Training uses object-pose offsets; end-effector offsets test perturbation-family transfer.

The first 15-seed run showed that boundary-heavy BGR-Suffix undercovered high-radius recovery. We therefore treat the coverage-aware variant as a separate manipulation-style support result, not evidence that boundary-only replay is robust: it adds broad-radius and uniform-state coverage and is promoted only after 30-seed confirmation, held-out replication, and stress sweeps. In the 30-seed run, BGR-Coverage raises final object RAUC over uniform by a small but consistent amount (0.4969 vs. 0.4854), and separates from failure-only (0.4229), clean-only (0.2631), loss-priority (0.1694), and fixed-radius replay (0.1583). The same run gives clean success 0.8644 vs. 0.8364, transfer RAUC 0.3143 vs. 0.3083, and AULC 0.3825 vs. 0.3709 over uniform. A held-out seeds 30–59 replication gives 0.4972 vs. 0.4859 RAUC, also with 30/0 wins. Pooling original and held-out suffix sweeps gives object RAUC 0.4971 vs. 0.4856, clean 0.8642 vs. 0.8367, transfer RAUC 0.3150 vs. 0.3089, and AULC 0.3829 vs. 0.3717. A 30-seed strategy ablation gives 0.4969 vs. 0.4587 and 0.4850 final object RAUC for coverage-aware, boundary-heavy, and hard-radius BGR; the hard-radius variant leads transfer RAUC and AULC (0.3252, 0.3928), but only coverage-aware replay improves final object RAUC over uniform with 30/0 wins.

A four-condition 30-seed suffix stress sweep varies teacher quality, clutter, feasible radii, and boundary sharpness. BGR-Coverage keeps 30/0 paired wins over uniform on clean success, final object RAUC, end-effector transfer RAUC, and AULC in every stress case. Across cases, BGR-Coverage gives object RAUC 0.3806–0.4971 vs. uniform 0.3755–0.4853 and transfer RAUC 0.2267–0.3142 vs. 0.2205–0.3081. Clean-only has a negligible clean-success edge (0.8646 vs. 0.8644), uniform remains higher on median r_{80} (0.5048 vs. 0.4985) pooled and in the original sweep (0.5047 vs. 0.4982), and uniform also has the stress-sweep median-r_{80} edge. We therefore treat suffix results as manipulation-style evidence, not final robotics evidence.

We audited the real-simulator path as premise validation. A GPU/EGL LIBERO-Goal probe completed 75/75 valid radius rows over five tasks, three reset states, five object-perturbation radii, and four trials per radius. Fixed-policy OpenVLA audits show clean replay success of 1.0, recovery drops under occlusion, blur, and shift, and BGR-boundary selection in the observed counterfactual-failure band $[0.25, 0.75]$ slightly more often than random-balanced selection (0.625 vs. 0.575).

The OpenVLA-OFT bridge exports 11,776 teacher-action rows; action-label/TFDS plumbing validates 2,048-transition matched BGR/random exports with 7D actions and 8D state, stock loader ingestion, and matched 10-step LoRA checkpoint smokes. These audits validate the interface but do not support a robotics success claim. A 10-task full-goal identity audit gives 99/100 clean successes for BGR, matched random, and the official checkpoint; the visual perturbation audit gives BGR 367/400 perturbed successes, tying official and trailing matched random by one episode (368/400). A 300-step image-augmentation continuation gives BGR and matched random 368/400 perturbed successes each, only one episode above official (367/400), while BGR trails both on identity. A 1,000-step low-learning-rate continuation is also negative: BGR gives 366/400 non-identity perturbation successes, trailing official at 367/400 and matched random at 370/400. The weighted perturbation curriculum is complete and also negative: BGR and official tie at 367/400 non-identity successes while matched random reaches 370/400. A proximal-anchor objective keeps BGR and matched random tied at 368/400 non-identity successes, only one episode above official (367/400), with BGR identity at 98/100 versus official 99/100. The perturb-only anchored objective preserves identity for all three methods at 99/100 and raises BGR to 371/400 non-identity successes, but official is 367/400 and matched random is 372/400, so BGR still fails the fixed +10/400 and 0.02 gate by trailing the matched control. Across these runs there is no stable official improvement and no stable learned-policy improvement over both comparators.

6 Limitations

BGR requires reset or replay access, a meaningful perturbation family, a feasibility witness, and extra rollouts for boundary estimation. It targets an observed boundary, not a formal robustness certificate. The clearest evidence is controlled procedural margin expansion; the synthetic and grid-margin benchmarks are constructed to expose recovery curves, so they establish mechanism rather than broad benchmark dominance. The paper does not claim a clear win on an independent pre-existing robustness benchmark.

Audit	BGR-family result	Blocking evidence	Decision
FrozenLake8x8	BGR 0.5453; clean 0.5581	uniform 0.5312; signs 14/16, 13/17	not promoted
MiniGrid FourRooms	BGR-Coverage 0.6077	uniform 0.6665; failure-only 0.7940; fixed 0.7587	negative
FourRooms mid2–5	BGR 0.6747; BGR-Cov. 0.5933	fixed 0.6779; failure-only 0.7309; r80 0.5627 vs. 0.6451	not promoted
DoorKey-6x6	BGR-Coverage 0.4846; BGR 0.3687	uniform 0.6384; failure-only 0.6459	negative
LavaCrossingS9N3	BGR-Coverage 0.3547; BGR 0.3153	uniform 0.4165; lower abs. radius	negative
LavaGapS7	BGR-Coverage 0.4277; BGR 0.4031	uniform 0.4461; uniform-radius 0.4627	negative
PointMaze U-Maze	BGR-Clean-Shield 0.2448; BGR 0.1406	failure-only 0.5458; 2/4 wins; r20 0.1167 vs. 0.2500	not promoted
FetchReach-v4	BGR-Coverage 0.4625; BGR 0.4438	uniform 0.6813; failure-only 0.8250; TD-loss 0.9437	negative
LunarLander-v3	BGR-Coverage 0.7500	uniform 0.6222; only 2/4 wins; r80 0.4200 vs. 0.4825	not promoted
bsuite/MinAtar	Catch BGR 0.8446; Cartpole BGR-Cov. 0.7559; MinAtar BGR 0.8896	Catch uniform 0.8782; Cartpole TD-loss 0.7694; MinAtar ties uniform	negative
Reacher-v5	BGR 0.2907; BGR-Coverage 0.2721	uniform 0.3862; BGR wins 4/12 seeds	negative
InvertedPendulum-v5	BGR 0.7500; BGR-Cov. 0.7500	uniform/fixed/failure-only/TD-loss all 0.7500; 0/0/4 ties	tied negative
InvertedDoublePendulum-v5	BGR 0.0833; BGR-Cov. 0.0000	uniform 0.0035; clean collapse 0.2500/0.0000	negative

Table 8: Standard-environment scope audits (limitations, not positive claims).

Table 8 keeps the scope explicit. FrozenLake has a small BGR RAUC edge but loses paired-sign and complementary-metric checks; MiniGrid, PointMaze, FetchReach, Reacher, bsuite, and MinAtar fail against stronger baselines, radius checks, or paired wins; and InvertedPendulum ties all methods. LunarLander is a near miss: BGR-Coverage has the best mean RAUC (0.7500 vs. 0.6222 uniform) but wins only 2/4 seeds and has lower median r_{80}. bsuite Catch shows scale-up fragility (0.8446 vs. 0.8782 uniform), bsuite Cartpole trails TD-loss replay, and MinAtar Breakout ties uniform and failure-only at 0.8896 RAUC with saturated $r_{80} = 5.0000$. InvertedDoublePendulum-v5 gives a tiny BGR RAUC edge (0.0833 vs. 0.0035 uniform) but collapses clean success to 0.2500 for BGR and 0.0000 for BGR-Coverage. FourRooms radius-10, HandReach-v3, and highway-fast-v0 follow-ups close simple rescue paths with saturated or invalid recovery curves. Other local tabular and classic-control probes remain internal diagnostics rather than manuscript evidence.

The suffix simulator is positive on final RAUC, clean success, transfer RAUC, and AULC, including stress regimes, but gains are small and uniform retains the median-critical-radius edge. RAUC and AULC are useful for measuring curve expansion, but they are author-defined integrals over chosen radius grids; we report median r_{80} and clean success separately instead of treating RAUC alone as decisive. OpenVLA is a learned-policy audit rather than a robotics training claim: it validates recovery-curve measurement and matched action/TFDS construction, while showing that a full robotics claim still requires stable optimization, clean-data scale, and LIBERO gains over both matched random and the official checkpoint.

7 Conclusion

BGR converts recovery-margin measurement into a replay curriculum. With reset access, feasible perturbations, and directly trainable recovery curves, boundary replay improves margin expansion over uniform, fixed-radius, failure-only, and loss-proxy replay; hard-state priority alone loses the grid-margin effect when radii are drawn uniformly. Synthetic and grid-margin studies establish the mechanism; suffix results show scoped manipulation-style transfer. The current learned-policy evidence remains an audit, not the final robotics result.

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Reproducibility Checklist

This checklist is filled for the current BGR submission package. The main paper is anonymous; commands, configs, per-seed CSVs, and run ledgers are in the anonymous artifact's `README.md` and `results/README.md`.

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1. General Paper Structure

1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA)

yes. The method section defines replayable states and recovery curves. It also states critical radii, BGR priority, and a training-loop method box.

1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no)

yes. The paper separates controlled margin-expansion claims, robot-suffix simulator results, OpenVLA fixed-policy audits, OpenVLA-OFT adaptation diagnostics, and tabular grid-policy setup failures.

1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no)

yes. The related-work section cites curriculum learning, PLR, PAIRED/ACCEL, domain randomization, MiniGrid, LIBERO, and OpenVLA.

2. Theoretical Contributions

2.1. Does this paper make theoretical contributions? (yes/no)

yes. The paper introduces a recovery-margin formalism and states a local margin-update proposition for boundary-centered radius sampling.

If yes, please address the following points:

2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no)

yes. The local boundary-intuition calculation states the local update-kernel and stochastic-dominance assumptions; the limitations section states replay-access and perturbation-family restrictions.

2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no)

yes. The local theoretical intuition is stated as a design rationale. Empirical claims are stated with measured metrics and paired-seed tests.

2.4. Proofs of all novel claims are included (yes/partial/no)

yes. The local boundary-intuition calculation includes a proof sketch based on monotone coupling and monotonicity of the update kernel.

2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no)

yes. The method section gives the local margin-update intuition and the proposition proof sketch.

2.6. Appropriate citations to theoretical tools used are given (yes/partial/no)

yes. The proof cites a standard stochastic-orders reference for the quantile coupling used in the local boundary-intuition calculation.

2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA)

partial. The proposition is a local model; the grid ablation empirically tests its main implication that boundary-centered radius sampling matters.

2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA)

yes. Ablation scripts, configs, per-seed CSVs, and claim-checking scripts are included.

3. Dataset Usage

3.1. Does this paper rely on one or more datasets? (yes/no)

partial. Main BGR training results use procedural synthetic/grid/suffix generators. The LIBERO/OpenVLA recovery audit uses existing benchmark task definitions, init states, and fixed-policy rollout artifacts to validate learned-policy recovery-curve measurement.

If yes, please address the following points:

3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA)

yes. The paper motivates synthetic recovery curves, procedural grid-margin replay states, robot suffix diagnostics, and LIBERO/OpenVLA as a realistic manipulation-policy target.

3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA)

yes. The experiments are generated by artifact-provided scripts/configs; per-seed result CSVs are included under `results/`.

3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA)

yes. The generated procedural data and code are included in the anonymous artifact under an MIT license.

3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA)

yes. LIBERO and OpenVLA are cited; MiniGrid is cited for grid-style context.

3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA)

yes for LIBERO task definitions, benchmark assets, and OpenVLA model code/weights. The anonymous artifact contains compact derived recovery summaries; raw fixed-policy rollout logs are retained for audit but are not required for the claimed quantitative results.

3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing (yes/partial/no/NA)

NA. No non-public dataset is used for claimed quantitative results.

4. Computational Experiments

4.1. Does this paper include computational experiments? (yes/no)

yes.

If yes, please address the following points:

4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA)

partial. Final configs and ablation variants are included in the anonymous artifact; the full search history is not exhaustively enumerated in the main paper.

4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no)

yes. Procedural generation, LIBERO probing, OpenVLA recovery summarization, aggregation, and figure scripts are included.

4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no)

yes. Experiment scripts, configs, tests, aggregation, and result CSVs are included in the anonymous artifact.

4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no)

yes. The anonymous artifact includes an MIT license, and the code plus generated procedural data will be made publicly available upon publication.

4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no)

partial. Code is modular and tested; comments are intentionally sparse and do not yet cross-reference paper sections.

4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA)

yes. Every config lists seeds; scripts pass seeds to NumPy generators and result CSVs include per-seed rows.

4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no)

yes. The run ledger records Slurm partitions, CPU/memory/time requests, hostnames for logged runs, environment constraints, and JSON environment snapshots for compute and GPU jobs.

4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no)

yes. Recovery AUC, critical radius, clean success, transfer RAUC, and AULC are defined or described.

4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no)

yes. Runs are reported in the paper and run ledger: synthetic training, estimator validation, grid-margin full-baseline, grid ablation, grid sensitivity confirmations, and robot-suffix coverage comparisons use 30 paired seeds; grid learning-curve diagnostics use 15 seed pairs. Smaller three- or five-seed exploratory diagnostics are labeled as diagnostics in the run ledger rather than used as main claims.

4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no)

yes. Aggregated tables include standard errors; experiments report median critical radius and AULC in addition to final means.

4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no)

yes. The anonymous artifact includes paired exact sign tests for the reported paired comparisons, including the 30-seed suffix coverage confirmation, grid-margin target-sensitivity, grid-margin obstacle-regime sensitivity, grid-margin geometry-stress sensitivity, and grid-margin learning-curve sweeps. The paper separates these main comparisons from smaller exploratory evidence.

4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA)

yes. Final hyperparameters are in the artifact YAML configs under `configs/`.